

NAME:TANKISO MASOEBE

COMPUTER ARCHITECTURE WK9

1.Scenario Analysis: Data Warehouse Setup for an E-Commerce Platform

An e-commerce platform can use a data warehouse to collect, organize, and analyze large amounts of business information from different systems. The data warehouse would receive data from sources such as customer accounts, product catalogs, online orders, payments, deliveries, and website activity. An ETL (Extract, Transform, Load) process would extract the data from these sources, clean and transform it, and then load it into the warehouse. The warehouse could contain fact tables for sales and orders, supported by dimension tables such as customers, products, dates, and locations. Managers could use business intelligence tools to create dashboards and reports showing sales trends, customer behavior, popular products, and revenue. Historical data would also help the company compare performance over different periods. This setup improves decision-making because employees can access accurate and centralized information without affecting the performance of the main e-commerce system.

2.Concept Research: Star vs. Snowflake Schemas

Star and snowflake schemas are data warehouse designs used to organize information for analysis and reporting. A star schema has a central fact table connected directly to dimension tables such as Customer, Product, and Date. It is simple, requires fewer joins, and usually provides faster queries, making it suitable for business intelligence and reporting. A snowflake schema extends the star design by normalizing dimension tables into additional related tables. This reduces data duplication and can improve storage efficiency and data organization, but it requires more joins and can make queries more complex. Star schemas are commonly preferred for simple, fast analysis, while snowflake schemas are useful for complex data hierarchies and reducing redundancy.

3.TOOL PRACTICE:Simulating an ETL Process in Python

I simulated an ETL process in Python to understand how data is prepared for a data warehouse. The process involved three main steps: extracting data from a source, transforming it by removing missing values and changing data formats, and loading the cleaned data into a destination file. Python libraries such as Pandas made it easier to manipulate and organize the data. The simulation showed how ETL improves data quality and ensures that information is consistent before analysis. I learned that ETL is useful for combining data from different sources and preparing it for reporting. This practice improved my understanding of data integration.

4. Application Practice: Data Warehouse Architecture

A data warehouse architecture can be designed with four main layers: data sources, ETL processing, the data warehouse, and reporting tools. Data sources include databases, e-commerce systems, customer records, and transaction systems. The ETL layer extracts data, cleans it, transforms it, and loads it into the central data warehouse. The warehouse stores organized historical data using fact and dimension tables. A data mart can also be created for specific departments such as sales or finance. Finally, reporting tools such as Power BI or Tableau allow users to create dashboards and analyze information. This architecture provides reliable data for decision-making, performance monitoring, and business planning.

